MIL-T-60500A(MU)
22 September 1970
SUPERSEDING
MIL-R-60500(MU)
30 September 1966

MILITARY SPECIFICATION

TRANSCEIVER: 10540501

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE EXERCISED IN OPERATING AND TESTING THE TRANSCEIVER. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE AND BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL, PAINT, ETC. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE, THE RANGEFINDER SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

1. SCOPE

1.1 This specification covers the transmitter-receiver assembly, hereinafter referred to as the transceiver, which is a component of the Rangefinder, Laser, AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-P-60290	Photomultiplier Power Supply: 10540714
MIL-Y-60291	Yoke Assembly: 10540506
MIL-C-60297	Counter, Time Interval: 10547200
MIL-N-60501	Nixie Power Supply: 10540716

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MIL-S-60502	Stop Preamplidier: 10540715
MIL-P-60503	Pulse Forming Network: 10540707
MIL-P-60504	Printed Circuit Board Assembly - Q Switch Timing Circuit: 10541270
MIL-C-60505	Cavity Assembly: 11727549
MIL-T-60506	Telescope Assembly: 10540506

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

D10541240	Chart, Boresight: 10541240
F8565556	Pressure Tester-Two Station
F10540501	Transceiver: 10540501
F10541111	Adapter Mount: 10541111
F11738155	Adapter, Vibration and Shock

PACKAGING DATA SHEET

10540501	Packaging of Transceiver 10540501

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The transceiver shall be manufactured in accordance with Drawing F10540501 and drawings pertaining thereto. The following

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components shall each have met the requirements of their detailed specification prior to assembly as a transceiver.

Telescope Assembly: 10540506	MIL-T-60506
Yoke Assembly: 10540507	MIL-Y-60291
Printed Circuit Board Assembly (Q Switch Timing): 10541270)	MIL-P-60504
Stop Preamplifier: 10540715	MIL-S-60502
Pulse Forming Network: 10540707	MIL-P-60503
Nixie Power Supply: 10540716	MIL-N-60501
Photomultiplier Power Supply: 10540714	MIL-P-60290
Cavity Assembly: 11727549	MIL-C-60505
Time Interval Counter: 10547200	MIL-C-60297

3.3 General specification.- The transceiver shall meet the following requirements, where applicable, of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Electrical and electronic assemblies
d. Soldering electrical connections
e. Part identification and marking
f. Workmanship

3.4 Reliability.- The transceiver shall exhibit a mean range between failure (MRBF) of 1700 ranges at a 95% confidence level.

3.5 Environmental.

3.5.1 Vibration.- The transceiver shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second (cps). The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequencies. Subsequent to vibration, the transceiver shall show no evidence of damage and shall meet the requirements of 3.5.9 through 3.7.10.2.

3.5.2 Shock.- The transceiver shall withstand a total of 9 shock impulses, 3 in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 milliseconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, there shall be no evidence of damage and the transceiver shall meet the requirements of 3.5.9 through 3.7.10.2.

3.5.3 Storage temperatures.- The transceiver shall be capable of withstanding temperature stabilization at +160°F and -80°F for a period of 5 hours at each temperature extreme. Subsequent to exposure to the temperatures specified herein, the transceiver shall show no evidence of damage and shall meet the requirements of 3.5.9 through 3.7.10.2.

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3.5.4 Humidity.- The transceiver shall be capable of withstanding 4 hours of continuous exposure to a relative humidity of 90 to 95 percent at a temperature of +40° ± 2°C. During and subsequent to the above humidity, the transceiver shall show no evidence of damage, and shall meet the requirements of 3.5.9 through 3.7.10.2.

3.5.5 Sand and dust.- The transceiver shall be capable of withstanding 3 hours of continuous exposure to blowing sand of particle size 0.01 to 1.00 mm and blowing dust of particle size 0.001 to 0.01 mm with a density between 0.1 to 0.5 gram per cubic foot of sand and a density between 0.1 to 0.5 gram per dubic foot of dust at a temperature of +25°C, a relative humidity between 15 to 30%, and a minimum wind velocity of 1500 ± 100 feet per minute. Subsequent to sand and dust exposure, the transceiver shall show no evidence of damage and shall meet the requirements of 3.5.9 through 3.7.10.2.

3.5.6 Salt spray.- The transceiver shall be capable of withstanding 48 hours of continuous exposure to a fine spray of 5% salt solution concentration with a specific gravity of from 1.0268 to 1.0413, a PH between 6.5 and 7.2, and a temperature between +93°F and +97°F. Subsequent to salt spray, the transceiver shall not have been damaged or adversely affected and shall meet the requirements of 3.5.9 through 3.7.10.2.

3.5.7 Fungus.- The transceiver shall be capable of withstanding 28 days exposure to a composite spore suspension consisting of the following cultures: (a) Chaetomium globosum, (b) Aspergillus versicalor, (c) Aspergillus flavus, (d) Pencillium luteum, (e) Rhizopus nigricans. The composite spore suspension shall be incubated at an ambient temperature of between +77°F and +84°F with an RH of 97 ± 2%. Subsequent to the above fungus exposure, the transceiver shall show no evidence of damage and shall meet the requirements of 3.5.9 through 3.7.10.2.

3.5.8 Rain.- The transceiver shall be capable of withstanding 5 hours of continuous exposure to heavy rainfall in the amount of 2.0 inches/hour with a wind speed of 35 knots at a water temperature of +85°F. Subsequent to such rainfall, the transceiver shall have no evidence of water penetration inside the volume housing, the electrical circuitry shall not have been damaged or adversely affected and the transceiver shall meet the requirements of 3.5.9 through 3.7.10.2.

3.5.9 Operational temperature.- Unless otherwise specified, the transceiver shall meet the requirements of 3.7 through 3.7.10.2 when thermally stabilized for a period of 2 hours at all temperatures from -65°F to +155°F inclusive.

3.6 Sealing.- The common area of the left, center and right housing of the transceiver shall be flushed with dry nitrogen. The nitrogen entering the right housing and bled from the left housing, until the gaseous mixture bled from the transceiver has a dew point at least as low

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as -25°F. The interior of the common area of the transceiver shall be capable of maintaining 5 pounds per square inch gage pressure for a period of at least 2 hours at a temperature of 75° ± 2°F over which time the transceiver leakage shall not exceed 0.2 psig. Subsequently, the internal pressure shall be reduced to normal atmosphere pressure and the sealing plugs placed in the ends of each housing. The interior atmosphere of the accepted transceiver shall be nitrogen with a dew point of -25°F.

3.7 Performance.- Unless otherwise specified, the following requirements shall be met for all temperatures from -65° to +155°F inclusive. With the transceiver mounted to a Yoke Assembly (10540507) both of which are mounted to a fixture representative of the Adapter Mount (10541111) and the transceiver energized through the electrical contacts of the Yoke Assembly from direct current sources capable of supplying the following voltages and currents,

a.	18 to 30 volts dc (nominal 24VDC)	5 amps max.
b.	7 to 9 volts dc (nominal 9VDC)	1 amp max.

the transceiver shall meet the requirements of 3.7.1 through 3.7.10.2. The transceiver shall meet the requirements when energized with all combinations of extreme voltages specified above.

3.7.1 Power consumption.

a. The current required by the transceiver from a nominal 24 VDC supply when the reticle lamp control is in the OFF position shall not exceed 60 milliamps in the quiescent state and shall not exceed 4 amperes at any time during operational cycle.

b. The current drawn by the transceiver from the nominal 9 VDC supply shall not exceed 70 milliamps in the quiescent state and shall not exceed 650 milliamps at any time during operational cycle.

NOTE: Operational cycle is the sequence of charging and firing necessary to range to a target.

3.7.2 Transmitter.

3.7.2.1 Power output.- At the Vo setting, adjusted for capacitance differences at assembly, the power output of the transceiver shall be no less than 1.9 MW when thermally stabilized at 75° ± 2°F.

3.7.2.2 Pulse shape.- The rise time of the output pulse meeting the requirements of 3.7.2.1 at +75°F ± 2°F shall not exceed 15 nanoseconds when measured between the ten percent and ninety percent levels. The width of the pulse shall not exceed 50 nanoseconds when measured at the 50 percent peak power points.

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3.7.2.3 Power output subsequent to all tests.- Subsequent to completion of all tests of this specification, the power output of the transceiver shall be at least 1.9 Mw when the transceiver is thermally stabilized at 75° ± 2°F and the external potentiometer is at the 0 position or 1 increment above the 0 position.

3.7.3 Receiver.

3.7.3.1 Receiver sensitivity.

a. The transceiver shall respond to signals having a power density at the receiver aperture of 25 microwatts/m^2 to 28 microwatts/m^2 at a wavelength of 6943 angstroms during the time interval between 10 and 60 microseconds after the laser pulse is emitted from the transceiver.

b. The transceiver shall not respond to signals of 20 microwatts/m^2 or less at a wavelength of 6943 angstroms during the time interval between 10 and 60 microseconds after the laser pulse is emitted from the transceiver.

3.7.3.2 Receiver noise.- With a bright background viewed through the telescope, the parallax shield down, the power switch in "ON" position, the read switch and the test switch depressed, simultaneously the 1 to 5 neon target discriminator lamp shall light and extinguish in sequence at an average rate of once per second.

3.7.4 Ranging accuracy.- The transceiver shall be capable of indicating the range to targets, located between 230 meters and 5,000 meters inclusive when local visibility conditions are 7500m or greater, and shall be capable of indicating the range to the targets to within ±10 meters of the actual range.

3.7.4.1 Range resolution.- The transceiver shall distinguish between targets separated by 30 meters in range and shall indicate the range to the first target, then the range to the second target.

3.7.4.2 Target discrimination.- The transceiver shall indicate the presence of 1, 2, 3, 4 and 5 or more targets visible within the boresight square of the telescope reticle when these targets are separated by a range of at least 20 meters but less than 45 meters and are beyond the selected setting of the minimum range switch. The transceiver shall also indicate the range to the target corresponding to the selected position of the target discriminator switch upon the same firing. The indication of the number of targets shall be displayed by the lighting of the appropriate target discriminator lamp when the read switch is depressed.

3.7.4.3 Minimum range selection.- With the minimum range selector switch rotated to each of the positions indicated in Column I, the transceiver, when fired at a target at the range indicated in Column II, shall not record a range reading on the nixie readout tubes. When the transceiver is fired at a range within the values indicated in Column III, the transceiver shall indicate the range within the requirements of range accuracy (see 3.7.4).

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I Selector Position	II Range	III Range
.2 = 200 meters	185 meter	230 meter
.6 = 600 "	585 "	625 "
1 = 1000 "	985 "	1025 "
2 = 2000 "	1985 "	2025 "
3 = 3000 "	2985 "	3025 "

3.7.4.4 Rapid ranging.- The transceiver shall indicate the range to a target located at 1500 meters when fired at 15 second intervals for 5 consecutive firings, followed by 10 consecutive firings at 1 minute intervals.

3.7.4.5 Delay ranging.- At a temperature between +70°F and +80°F, the transceiver shall indicate the correct range to a target at approximately 5,000 meters to within the range accuracy when the range switch is depressed no less than 75 seconds after the charge switch has been depressed.

3.7.4.6 Minimum time between rangings.- The transceiver shall be capable of cycling through a charging and firing sequence within 10 seconds.

3.7.5 Electronic display.

3.7.5.1 Power switch turn-on.- With the power switch initially placed in the "ON" position and the read switch depressed, the transceiver shall read four zeros at the nixie tubes with the number seven superimposed on the zero in the unit column and no target indicated (no light) on the five neon target discriminators.

3.7.5.2 Time interval counter test.- With the read switch depressed subsequent to depressing the test switch, the counter shall indicate a reading on the numerical indicators of between 20 and 50 meters.

3.7.5.3 Range counter reset.- The counter shall be reset by placing the power switch in the "ON" position and also when the RANGE switch is depressed 10 seconds or more after the CHARGE switch has been depressed. Indication that the counter is reset shall be given by 4 zeros with the number 7 superimposed on the zero in the unit column of the nixie readout when the READ switch is depressed.

3.7.5.4 Long range reading.- The transceiver shall demonstrate that it is capable of recording ranges out to 19,995 meters. When range is read on the nixie tubes, the number 1 shall be superimposed on the number that appears in the units column when ranging to a target in excess of 9999 meters

EXAMPLE - Numbers that appear on the 4 nixie tubes with a "0" in the units column, and are known to be targets beyond 1000 meters, indicate a target in the 10,000m range e.g. 0735 indicates a range reading of 10,735 meters.

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3.7.5.5 TGT discriminator lamps.- No more than 1 target discriminator lamp shall light to indicate the presence of multiple targets.

3.7.5.6 Low voltage indicator.- With the power switch in the ON position and the READ switch depressed, the numeral 2 shall appear in the unit column in addition to Numbers 0, 1, 5 or 7 whenever the nominal input voltages are less than 21 ± 1 volts or less than 7.2 ± 0.2 VDC respectively.

3.7.6 Elevation.

3.7.6.1 Illumination.- Depression of the switch on the yoke assembly shall illuminate the elevation and azimuth scales, and a circular level vial on the representative adapter mount. The intensity of the reticle light shall be smoothly variable from maximum brightness to completely off by turning the reticle adjustment potentiometer knob on the cover plate of the telescope assembly in the counterclockwise direction.

3.7.6.2 Elevation accuracy.- With the horizontal reticle line of the transceiver in coincidence with a horizontal target line, and the elevation scale and micrometer each in coincidence with their respective index, the scale and micrometer shall indicate the true position of the line of sight in elevation and depression including backlash, within 2.0 mils at elevation and depression readings of 75, 150, 300 and 500 mils.

3.7.6.3 Elevation backlash.- Backlash in the elevation mechanism at any point over the full excursion of travel 500 mils elevation and 500 mils depression from a horizontal line of sight shall not exceed 0.8 mil.

3.7.6.4 Elevation knob torque.- The running torque required to rotate the elevation knob in each direction of clockwise and counterclockwise rotation shall not exceed the values specified at temperatures indicated.

-65°F	counterclockwise	5 inch pounds
	clockwise	6.5 inch pounds
70° to 80°F	counterclockwise	25 inch ounces
	clockwise	40 inch ounces
+155°F	counterclockwise	25 inch ounces
	clockwise	40 inch ounces

3.7.6.5 Lateral displacement.- Any horizontal movement of the line of sight caused by reversing direction of rotation of the elevation knob shall not exceed 0.10 mil at any point throughout the excursion of plus and minus 500 mils.

3.7.7 Sighting.

3.7.7.1 Open sight.- The line of sight through the open sight shall be parallel within 20 mils in azimuth and elevation to the established line of sight of the sighting optical system.

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3.7.7.2 Collimation (boresight).- Using Chart, Boresight, 10541240 made in accordance with Drawing D10541240, positioned on a horizontal line of sight at a known distance of between 100 to 150 feet from the transceiver, the transceiver shall be oriented and fired with parallax shield closed to position the laser beam in the center of the rectangle on the target. With the transceiver at this position, through boresight adjustments, the horizontal and vertical reticle lines shall be positioned in coincidence with the horizontal and vertical target lines. The elevation scale on the telescope assembly and the dial scales of the micrometer shall each be adjusted to coincide with their respective zero indexes and locked in position.

3.7.7.3 Boresight adjustment.- The maximum torque required for rotation of the azimuth and elevation boresight screws shall not exceed 3 inch-pounds when the telescope assembly is thermally stabilized at 75° ± 2°F and shall not exceed 5 inch-pounds when thermally stabilized at the temperature extremes of 3.5.9.

3.7.7.4 Plumb travel.- With the elevation scale of the transceiver indicating 500 mils elevation, the line of sight through the center of the boresight square of the reticle shall be brought into coincidence with a plumb line. The line of sight through the reticle shall track the plumb line within ±1.0 mil (total excursion of deviation not to exceed 1.0 mil) when the transceiver line of sight is depressed over an excursion of 1000 mils by rotation of the elevation knob.

3.7.7.5 Eyepiece focus.- The diopter scale shall indicate zero diopters within plus or minus 1/4 diopter when the eyepiece focus is adjusted to zero diopters.

3.7.7.6 Diopter adjustment.- The maximum torque required for rotation of the diopter adjustment for eyepiece focus shall not exceed 10 inch-ounces when the transceiver is thermally stabilized at 75° ± 5°F and shall not exceed 15 inch-ounces when thermally stabilized at the temperature extremes of 3.5.9.

3.7.7.7 Reticle illumination.- With the reticle lamp energized with 28 volts, the reticle pattern shall be clearly defined when viewed in a darkened area. Rotation of the potentiometer knob clockwise from the extreme counterclockwise position shall complete the electrical circuit and then increase the intensity of reticle illumination from minimum to maximum.

3.7.8 Electromagnetic compatibility.- The radiated and conducted electromagnetic emissions from the transceiver shall not exceed the allowable limits of MIL-STD-461 for Class IIA equipment.

3.7.9 Interchangeability.- The transceiver shall meet all the requirements of this specification without need for selective assembly of components/modules. They shall be interchangeable such that replacement of any single component/module or random combination of components/modules of the trans-

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ceiver permits the transceiver to operate and perform to the requirements of this specification. No incompatibility shall exist between the component/module replacements listed below and the mechanical interfaces, optical performance characteristics or electrical functions of the transceiver.

(1) Panel Assembly 10540701 with Counter 10547200
(2) Nixie Power Supply 10540716
(3) Stop Preamplifier 10540715
(4) Photomultiplier Tube 10540742
(5) Photomultiplier Power Supply 10540714
(6) Q-Switch Timing Circuit 10541270
(7) Trigger Module 10540989
(8) Voltage Divider 10541267
(9) Cavity 11727549
(10) PFN Power Supply 10540707
(11) Microscope 10540941
(12) Yoke Assembly 10540507
(13) Telescope Assembly 10540506
(14) Left Housing Assembly 10540969

3.7.10 Safety.

3.7.10.1 Operation.

a. When the power switch of the transceiver is in the "ON" position and the capacitor of the pulse forming network (PFN) is at full charge, placement of the power switch to the "OFF" position shall cause the PFN to discharge through the flash lamp and the motor shall not run.

b. When the power switch is in the "ON" position and the range switch is depressed, without depressing the "CHARGE" switch, the motor shall run but the flash lamp shall not fire.

c. When the capacitors of the PFN are at full charge, removal of the cover of the left housing shall cause the energy stored in the capacitors to discharge through an internal load resistor.

3.7.10.2 Eye filter.

a. The absorbing glass filter of the eye filter assembly (10558179) of the transceiver shall attenuate the laser light reflected to the eye by a factor of 100,000.

b. The filter lever of the eye filter assembly shall be operable under all environmental conditions and shall allow for removal of the filter from the line of sight of the transceiver when the lever is depressed and replacement of the filter into its original position into the line of sight of the transceiver when the lever is released.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 6 transceivers plus 6 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS
(100% INSPECTION)

		REQUIREMENT	TEST PROCEDURE
101.	Sealing	3.6	4.6.4
102.	Power consumption	3.7.1	4.6.5.1.1
103.	Power output	3.7.2.1	4.6.5.1.2
104.	Pulse shape	3.7.2.2	4.6.5.1.3
105.	Power output subsequent to all tests	3.7.2.3	4.6.5.1.4
106.	Receiver sensitivity	3.7.3.1	4.6.5.2.1
107.	Receiver noise	3.7.3.2	4.6.5.2.2
108.	Ranging accuracy	3.7.4	4.6.5.3
109.	Range resolution	3.7.4.1	4.6.5.3.1
110.	Target discrimination	3.7.4.2	4.6.5.3.2
111.	Minimum range selection	3.7.4.3	4.6.5.3.3
112.	Rapid ranging	3.7.4.4	4.6.5.3.4
113.	Delay ranging	3.7.4.5	4.6.5.3.5
114.	Minimum time between rangings	3.7.4.6	4.6.5.3.6
115.	Long range readings	3.7.5.4	4.6.5.4.1
116.	Power switch turn-on	3.7.5.1	4.6.5.4.2
117.	Time interval counter	3.7.5.2	4.6.5.4.3
118.	Range counter reset	3.7.5.3	4.6.5.4.4
119.	TGT discriminator lamps	3.7.5.5	4.6.5.4.5
120.	Low voltage indicator	3.7.5.6	4.6.5.4.6
121.	Illumination	3.7.6.1	4.6.5.5.1
122.	Elevation accuracy	3.7.6.2	4.6.5.5.2
123.	Elevation backlash	3.7.6.3	4.6.5.5.3
124.	Elevation knob torque	3.7.6.4	4.6.5.5.4
125.	Lateral displacement	3.7.6.5	4.6.5.5.5
126.	Open sight	3.7.7.1	4.6.5.6.1
127.	Collimation (boresight)	3.7.7.2	4.6.5.6.2

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128.	Boresight adjustment	3.7.7.3	4.6.5.6.3
129.	Plumb travel	3.7.7.4	4.6.5.6.4
130.	Eyepiece focus	3.7.7.5	4.6.5.6.5
131.	Diopter adjustment	3.7.7.6	4.6.5.6.6
132.	Reticle illumination	3.7.7.7	4.6.5.6.7
133.	Safety operation	3.7.10.1	4.6.5.9.1
134.	Eye filter	3.7.10.2.b	4.6.5.9.2
135.	Operational temperatures (-65°F, +75° ± 2°F and +155°F)	3.5.9	4.6.2.9

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristic specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Eye filter	3.7.10.2.a

4.4 Special sampling.

4.4.1 General.

4.4.1.1 Three transceivers shall be selected at random as a special sample from each 50 produced or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable drawings & specs - Visual
302.	Fabrication	3.2	Applicable drawings & specs - Visual
303.	General specification	3.3	MIL-F-13926 - Visual
304.	Electromagnetic compatibility	3.7.8	4.6.5.7

NOTE: The tests of Table II shall be conducted at a temperature between +70° and +80°F.

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4.4.1.2 Three transceiver components/modules shall be selected at random as a special sample from each month's production. Each sample shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
305.	Interchangeability	3.7.9	4.6.5.8

4.4.2 Environmental.- Three transceivers shall be selected at random as a special sampling from each 50 produced, or from each month's production, whichever occurs first. Each sample shall be subjected to the tests in Table IV and shall then meet the requirements and tests in Table I.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
306.	Storage temperature (-80° and +160°F)	3.5.3	4.6.2.1
307.	Shock (between +70° and +80°F)	3.5.2	4.6.2.2
308.	Vibration (between +70° and +80°F)	3.5.1	4.6.2.3
309.	Humidity	3.5.4	4.6.2.4
310.	Sand and dust	3.5.5	4.6.2.5
311.	Salt spray	3.5.6	4.6.2.6
312.	Rain	3.5.8	4.6.2.8

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Documents when provided with the technical data package (TDP).

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4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Vibration.- Special testing equipment utilized in the vibration test of the transceiver shall conform to Drawing F11738155.

4.5.2.3.2 Shock.- Special testing equipment utilized in the shock test of the transceiver shall conform to Drawing F11738155.

4.5.2.3.3 Sealing.- Special pressure testing equipment to test sealing of the transceiver shall conform to the internal pressure test requirements of Drawing F8565556.

4.5.2.3.4 Performance - transceiver assembly.- Special testing equipment to test the transceiver assembly shall conform to the following requirements:

a. Special testing equipment utilized in testing the transceiver shall consist of a mounting adapter with a stand or tripod. The adapter shall contain the necessary locating surfaces, rotating and locking features, and electrical circuitry of the Adapter Mount, Drawing F10541111 that are necessary to accept, secure and energize the yoke assembly of the transceiver. The mounting adapter shall contain an azimuth scale. The mounting adapter shall also contain a circular level vial and crossleveling screws for crossleveling the transceiver within 10 seconds of arc. A power supply for energizing the transceiver shall be part of the special testing equipment. The power supply shall be capable of supplying and maintaining the required voltages for tests. The input voltages and currents shall be monitored with appropriate voltmeters and ammeters for measuring the power consumption of the transceiver during all modes of operation.

b. Wall targets.- A calibrated wall target with appropriate target patterns for measuring the optical and mechanical accuracies of the transceiver specified in 4.6.5.5 shall conform to the following requirements:

(1) The target shall be positioned approximately 5 feet ±1/32 inch in front of the transceiver and aligned to the axis of rotation

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of the telescope assembly. The "0" horizontal line of sight of the target image shall be set in the same (leveled) horizontal plane as the "0" horizontal line of sight of the transceiver. The elevation and depression target images (lines of sight) shall be positioned in the same vertical plane with respect to the "0" horizontal line of sight throughout 1,000 mils to an accuracy of .1 of a mil. The target images shall be capable of measuring the accuracies specified in 4.6.5.5 when introduced into the transceiver at the positions corresponding to the elevation scale of the telescope and dial scale of micrometer. The target image shall be equipped with a "0" calibrated vertical line capable of measuring the accuracy of lateral displacement specified in 3.7.6.5. Tolerance marks shall also be incorporated on both sides of the "0" vertical line to determine plumb travel accuracies specified in 3.7.7.4 with respect to the "0" vertical target line.

c. Auxiliary diopter power.- When the transceiver is subjected to the tests in 4.6.5.5, the wall target may be outside the range of eye-piece adjustment. In this case, utilize an auxiliary diopter lens, between +.60 to +.65, positioned close to the sighting objective assembly of the transceiver in the same plane of the entrance pupil to obtain best focus setting of the target images. Upon completion of the set-up procedures, the parallax shield of the transceiver shall be secured in the closed detent position thereby reducing the amount of stray light entering the transceiver and parallax which may be present when performing the tests (4.6.5.5) at the target distance specified (see b.1).

4.5.2.3.5 Power output.- The power output of the transceiver shall be measured by a method referrable to NBS standards with an accuracy of .2 megawatts. The test set-up and equipment shown in Figures 1 and 2 is acceptable, but is not mandatory. Use of a non-specular (matte finish) target board of high reflectance (at least 85%), as shown in Figure 1 is recommended to attenuate the beam before incidence on the detector. Dimensions shown are approximate, for illustrative purposes only. The detector shall be calibrated by means of NBS standard(s) to measure the output power of the transceiver in the specific set-up employed. The method of power measurement, supported by appropriate calibration data, shall be Government approved prior to being utilized in the final acceptance inspection of the transceiver.

4.5.2.3.6 Receiver sensitivity.- Special test equipment to be utilized to perform the receiver sensitivity test shall be as specified in 3.7.3.1 and shown in Figure 3.

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a. Output detector.- The output detector shall be capable of detecting the output pulse from the transceiver as specified in 3.7.2.2.

b. Time delay circuit.- This circuit shall be capable of delaying the pulse from the output detector a minimum of 10 microseconds to a delay of 130 microseconds.

c. Pulse generator.- The pulse generator shall be capable of receiving the pulse from the time delay circuit. The external trigger delay shall be less than 20 nanoseconds between leading edge of external trigger input pulse and leading edge of output pulse. The output pulse width shall be at least 30 nanoseconds and be continuously variable to 100 nanoseconds.

d. Pulse light source.- This light source shall utilize a light emitting diode and a spectral filter. The filter shall be a band pass filter peaked at 6943Å ± 2Å with a minimum transmission of 50% and a one half band width of 10 ± 2Å. The L.E.D. shall produce a light pulse with a rise time not to exceed 15 nanoseconds when measured between the 10% and 90% amplitude levels. The pulse width shall not exceed 50 nanoseconds when measured between the 50% peak power points. The light emitting diode (LED) shall be capable of producing at the transceiver aperture of the rangefinder an irradiance of at least 20 microwatts/meters^2 and not more than 30 microwatts per square meter, and shall be capable of varying the irradiance for the range of 20-30 microwatts per square meter.

e. Irradiance detector.- This detector shall be capable of measuring from 20 ± 1 microwatts per square meter to 30 ± 1 microwatts per square meter from the pulse light source.

4.5.3 Test equipment.- In conjunction with 4.5.2.3, the following standard test equipment shall be utilized in performance of the applicable tests as specified in 4.6.

NOMENCLATURE	DESCRIPTION
1. Oscilloscope (CRO) with camera (see Figures 1 & 2)	Frequency response, DC to 85 megacycles; rise time 4.5 nanoseconds; input impedance 10,000 megohms or greater; sensitivity 50 mv/cm; sweep time. 1 microsecond per centimeter, no parallax graticule; camera lens f/1.9, 1.05 object to image ration; film ASA equivalent of 10,000 (Polaroid Type 410 or equivalent)
2. DC power supply(s)	0 to 30 volts, 5 amps; 0 to 9 volts, 1 amp; ripple less than 0.1 volt RMS, regulation 1%
3. Voltmeter D.C.	Range: 0 to 50 volts Accuracy: Within 2% of full scale reading
4. Ammeter D.C.	Range: 0 to 10 amps Accuracy: Within 2% of full scale reading

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4.6 Test methods and procedures.- The following limitations are recommended for the protection of personnel against potential hazards from backscattered laser radiation.

a. No tests shall be conducted in rain, snow or hail.

b. Testing at night or day twilight should be avoided; however, if necessary, all personnel using the rangefinder, or in the vicinity, shall be equipped with a laser safety glass of at least density 2 at 6943 angstroms.

c. Testing in daytime in heavy fog (visibility less than 200 meters) should be avoided; otherwise, safety glasses as specified above should be employed.

4.6.1 Fabrication.

4.6.1.1 Transceiver components.- The inspection records of each transceiver submitted for final acceptance shall show evidence that the components indicated in 3.2 conform to the requirements specified therein. The voltage marking (Vo voltage value) on the cavity assembly utilized in the transceiver shall be recorded on the inspection record of the transceiver along with the value of capacitance of the transceiver.

4.6.2 Environmental.- The transceiver shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the transceiver shall be subjected to the tests specified in 4.6.4 through 4.6.5.9.

4.6.2.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.5.3.

4.6.2.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the test equipment specified in 4.5.2.3.2. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the waveform, time duration and peak "G" value specified in 3.5.2.

4.6.2.3 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus specified in 4.5.2.3.1. The transceiver shall be firmly mounted to the vibration table. The vibration equipment shall provide the conditions of acceleration amplitude, frequency and duration specified in 3.5.1. The vibration test shall then be performed in accordance with Method 510 Procedure I, Part 1 of MIL-STD-810.

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4.6.2.4 Humidity.- The humidity test shall be performed as outlined in MIL-F-13926, Procedure II and under the conditions specified in 3.5.4 for 4 hours.

4.6.2.5 Sand and dust.- The sand and dust test shall be performed as outlined in MIL-F-13926, with the following conditions: The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.5.5.

4.6.2.6 Salt spray.- The salt spray test shall be performed as outlined in MIL-STD-810, Method 509, and according to the conditions specified in 3.5.6.

4.6.2.7 Fungus.- One (1) transceiver shall be subjected to the fungus test outlined in MIL-F-13926. At the end of the 28 day incubation period, the transceiver shall be tested as specified in 4.6.5 through 4.6.5.9 for compliance with 3.5.9 through 3.7.10.2. Subsequent acceptance of the transceiver shall be by certification.

4.6.2.8 Rain.- The rain test shall be performed as outlined in MIL-F-13926, Procedure I with the following exceptions: Five (5) hours of continuous rainfall at a rate of 2 inches/hour and wind velocity of 35 knots. Temperature of the water shall be maintained at approximately 85°F. Subsequent to the rainfall test, the requirements of 3.5.8 shall be met.

4.6.2.9 Operational temperatures.- Unless otherwise specified, the test shall be performed in accordance with Procedure I of MIL-F-13926 at the extreme temperatures specified in 3.5.9 and at a temperature of 75° ± 2°F utilizing the test equipment indicated in 4.6.2.1. The transceiver shall be exposed to each specified temperature for a minimum period of 2 hours. At the end of each 2 hour period, and while still at the specified temperature, the transceiver shall be subjected to the tests in 4.6.5 through 4.6.5.9.2 to determine conformance to the applicable requirements of 3.7 through 3.7.10.2.

4.6.3 Reliability.- Reliability testing is required to be performed in conjunction with First Article Testing 1/. The reliability sample shall consist of 6 transceivers with a minimum of 800 ranges per transceiver. During those firing tests outlined in this specification, additional firing shall be performed in arder that the transceiver reliability may be verified.

A record shall be kept for each transceiver indicating serial number, type of environmental exposure, number of failures (type and cause), and range data (true range and transceiver range), as well as number of targets within reticle of transceiver, the number of targets indicated by the target discriminator lamps and position of the target discriminator switch, type of power supply, rangings per battery serial number.

1/ When the system specification is part of the contract, the reliability tests associated with the system specification MIL-R-50608 will be performed in lieu of the reliability test of this specification.

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An entry shall be made into the record each time the transceiver is fired. The data collected will be reduced and compared to the following relationship:

$$1700 \leq \frac{2T}{\chi^2_{.05,\ 2(r+1)}}$$

where T = total number of rangings

r = number of system failures*

χ^2 = Chi Square

2(r+1) = degrees of freedom

4.6.4 Sealing.- The flushing and internal pressurizing of the transceiver shall be performed using dry nitrogen gas and special testing equipment conforming to 4.5.2.3.3. A dew point tester, with appropriate adapters shall be utilized in flushing and pressurizing the transceiver in accordance with the applicable requirements specified in 3.6. The transceiver shall be flushed with the dry nitrogen gas until the gaseous mixture bled from the transceiver conforms to the dew point requirements specified in 3.6. The flow of nitrogen gas into the transceiver during the flushing operation shall be regulated so as not to exceed 5 psig. Upon completion of the flushing operation, the purging screw shall be closed. The transceiver shall then be pressurized with dry nitrogen gas to 5 psig in accordance with the procedures for internal pressure testing of Drawing F8565556 at the temperature specified in 3.6. Subsequent to reaching the pressure specified, the source of pressure to the transceiver shall be shut off. At the conclusion of the specified time period, the leakage shall have not exceeded the requirements of 3.6 as indicated on the pressure gage of the testing equipment. Upon completion of the sealing test, the internal pressure, final sealing and interior atmosphere of the accepted transceiver shall conform to the requirements of 3.6.

4.6.5 Performance.- The following tests shall be conducted with the transceiver positioned on the special testing equipment specified in 4.5.2.3.4. Unless otherwise specified, the input voltages to the transceiver shall be established at a nominal 24.0 ± 2.0 volts d.c. at 5 amperes maximum, and 9.0 ± 0.5 volts d.c. at 1 ampere maximum.

NOTE: For the protection of test personnel, a safety shield should be utilized to cover the transmitter aperture of the telescope assembly during those tests that do not require transmission of the laser beam.

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*Transceiver failure: Any malfunction which renders the transceiver inoperative or operable outside the limits of the specification.

*Transceiver interruption: Any system malfunction which does not render the transceiver inoperable within the limits set forth in this specification but requires maintenance to correct the malfunction. Subsequent to the reliability test, the sample shall be refurbished and readied for delivery and shipment.

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4.6.5.1 Transmitter.

4.6.5.1.1 Power consumption.- The power consumption of the transceiver shall be measured by means of the applicable monitoring voltmeters and ammeters specified in 4.5.2.3.4. Amperage measurements shall be made during all modes of operation specified in 3.7.1. Power consumption of the transceiver during the modes of operation specified, shall conform to the applicable requirements of 3.7.1.

4.6.5.1.2 Power output.- The following tests shall be conducted with the transceiver positioned in a temperature test chamber that conforms to the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.7.2.1, 3.7.2.2 and 3.7.2.3. The test equipment specified in 4.5.2.3.5 and 4.5.3 utilized in this test shall be positioned externally, except the mounting adapter specified in 4.5.2.3.4, as shown in Figure 1. The necessary leads and connections to the transceiver shall be incorporated in the test chamber. The chamber shall be provided with an opening to permit firing of the transceiver as shown in Figure 1. After the transceiver has been thermally stabilized for a period of 2 hours, the transceiver shall be fired a minimum of 3 times. The power output shall meet the requirements of 3.7.2.1. A period of 10 minutes shall elapse between each firing. The average power output for 3 firings shall be computed and recorded.

4.6.5.1.3 Pulse shape.- Conduct this test in conjunction with the test specified in 4.6.5.1.2. The rise time and width of the pulse shall meet the requirements specified in 3.7.2.2.

4.6.5.1.4 Power output subsequent to all tests.- Position the external potentiometer of the transceiver as specified in 3.7.2.3. After the transceiver has been thermally stabilized for a period of 2 hours at 75° ± 2°F, and while still at the temperature specified, the transceiver shall be fired a minimum of 3 times. The power output shall meet the requirements of 3.7.2.3. A period of 10 minutes shall elapse between each firing. The average power output for 3 firings shall be computed and recorded.

4.6.5.2 Receiver.

4.6.5.2.1 Receiver sensitivity.- The special testing equipment specified in 4.5.2.3.6 shall be utilized in performing this test. The test equipment shall be positioned as showed in Figure 3. The special test equipment shall be calibrated and set-up as follows before each test. Place transceiver in position and connect the output detector over the transmitter aperture, and connect the irradiance detector over the receiver aperture. Align the light source so that the full intensity from the L.E.D. is directed to the aperture of the received. Connect and energize all the equipment as shown in Figure 3. Fire the transceiver and adjust the time delay between 10 microseconds and 60 microseconds and then adjust the irradiance at the receiver aperture for the value specified in 3.7.3.1.a. Remove the

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irradiance detector. Position the power switch to "ON" and fire the rangefinder. A range reading of between 1500 and 9000 meters displayed on the rangefinder counter shall indicate conformance to the requirements of 3.7.3.1.a. The above procedure shall be repeated with an irradiance at the receiver aperture of 20 microwatts per square meter. No range reading will indicate conformance to the requirements of 3.7.3.1.b.

4.6.5.2.2 Receiver noise.- Position an illuminated 100 watt frosted incandescant lamp on the optical axis of the sight receiver, at a distance of 9 feet ± 1 foot from the sight receiver objective lens. Position the power switch to "ON". Depress the read and test switches simultaneously and observe the 5 neon target discriminator lamps. The 5 neon target discriminator lamps shall conform to the requirements specified in 3.7.3.2. A standard stop watch shall be utilized for measuring the sequencing rate of the lighting and extinguishing of the lamps.

4.6.5.3 Ranging accuracy.- Perform this test with targets located at 230, 625, 1025, 2025, 3025, 4135, and 5000 meters from the transceiver. The measured distance to each target between 230 and 5000 meters shall be established within ±0.5 meters. The transceiver shall be fired at each target utilizing the appropriate setting of min-range switch, and the discriminator switch where necessary. Record the range reading to each target. The recorded range reading to each target shall conform to the requirements specified in 3.7.4.

4.6.5.3.1 Range resolution.- Perform this test utilizing 2 targets. Target 1 shall be positioned at 4500 meters. Target 2 shall be located at 30 meters + 0.2 meter behind Target 1. The 2 targets shall be clearly visible and be within the boresight square of the transceiver reticle, when viewed through the transceiver eyepiece assembly. Rotate the target discriminator switch to the number "1" position. The transceiver shall be fired to the targets. Record the range reading. Rotate the target discriminator switch to the number "2" position. The transceiver shall again be fired to the targets. Record the range reading. The range resolution of the transceiver shall meet the requirements specified in 3.7.4.1.

4.6.5.3.2 Target discrimination.- Perform this test with a group of 5 targets positioned between 250 and 550 meters. The group of targets shall be positioned as required in 3.7.4.2. The distance to each target shall be known. Position the min-range switch of the transceiver to the "2" position.

1. Position discriminator switch to the "1" position and perform the following operations:

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a. Adjust the orientation of the transceiver so only the first of the 5 targets is visible with the boresight square and fire the rangefinder. Depress the read switch and observe the first target discriminator lamp illuminated and the range to the target.

b. Adjust the orientation of the transceiver so that only the first and second target are visible within the boresight square and fire the transceiver. Depress the read switch and observe only the second target discriminator lamp illuminated and the range to the first target.

c. Adjust the orientation of the transceiver so that only the first 3 targets are visible within the boresight square and fire the transceiver. Depress the read switch and observe only the third target discriminator lamp illuminated and the range to the first target.

d. Adjust the orientation of the transceiver so that only the first 4 targets are visible within the boresight square and fire the transceiver. Depress the read switch and observe only the fourth target discriminator lamp illuminated and the range to the first target.

e. Adjust the orientation of the transceiver so that all 5 of the targets are visible within the boresight square and fire the transceiver. Depress the read switch and observe only the fifth target discriminator lamp illuminated and the range to the first target.

2. Place the target discriminator switch in the "2" position and repeat operations "1a" through "1e" above observing the same results except that the range observed shall be to the second target when it is visible with the boresight square and a no target range indication otherwise.

3. Place the target discriminator switch in the "3" position and repeat operations "1a" through "1e" above observing the same results except that the range observed shall be to the third target when it is visible within the boresight square and the no target range indication otherwise.

4. Place the target discriminator switch in the "4" position and repeat operations "1a" through "1e" above observing the same results except that the range observed shall be to the fourth target when it is visible within the boresight square and the no target range indication otherwise.

5. Place the target discriminator switch in the "5" position and repeat operations "1a" through "1e" above observing the same results except that the range observed shall be to the fifth target when it is visible within the boresight square and a no target range indication otherwise.

4.6.5.3.3 Minimum range selection.- Perform this test with the targets positioned at known distances as specified in 3.7.4.3, Columns II and III. With the minimum range selector switch rotated to each position indicated in Column 1, and the discriminator switch positioned to the number "1", the transceiver shall be fired at the applicable range indicated in

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Column II, there shall be no record of a range reading on the numerical readout. Subsequent to this test, the transceiver shall be fired to the target ranges specified in Column III. Record the range readings. Upon completion of all the ranging tests, the minimum range switch shall have met the requirements specified in 3.7.4.3, at the indicated ranges.

4.6.5.3.4 Rapid ranging.- Perform this test utilizing a target positioned at a known distance of 1500 meters ±1 meter. Orient the transceiver so that the target is clearly visible in the boresight square of the reticle. The transceiver shall be fired at 15 second intervals for 5 consecutive firings followed by 10 consecutive firings at 1 miniute intervals. The requirements specified in 3.7.4.4 shall be met.

4.6.5.3.5 Delay ranging.- Perform this test utilizing a target positioned at a known distance of 5000 ± 100 meters. Orient the transceiver so that the target is clearly visible in the boresight square of the reticle. Depress the charge switch, then after a delay of at least 75 seconds depress the range switch. The correct range to the target shall be indicated in numerical display as specified in 3.7.4.5.

4.6.5.3.6 Minimum time between rangings.- With the power switch in the "ON" position depress the charge switch, then depress the range switch. The time interval between the depression of the charge switch, and the audible popping noise of the discharging of the pulse forming network through the flash lamp shall conform to the requirements of 3.7.4.6, as measured with a standard stopwatch.

4.6.5.4 Electronic display.

4.6.5.4.1 Long range reading.- Utilizing the test methods of 4.6.5.2.1, except that the time delay shall be between 100 microseconds and 130 microseconds, the transceiver shall display a range reading between 15,000 meters and 19,995 meters.

4.6.5.4.2 Power switch turn-on.- Position the power switch to "ON". Depress the read switch. The four (4) numerical indicators and target discriminators shall conform to the requirements specified in 3.7.5.1.

4.6.5.4.3 Time interval counter.- Position the power switch to "ON". Depress the test switch, then depress the read switch while observing the numerical display. The counter shall conform to the requirements of 3.7.5.2.

4.6.5.4.4 Range counter reset.- Position the power switch to "ON". Depress the test switch, then depress the read switch while observing the numerical display. The counter shall conform to the requirements of 3.7.5.3. Depress the test switch then depress the read switch while observing the numerical display. A range less than 100 meters but greater than zero shall be observed. Depress the charge switch,allow ten seconds to elapse, then depress and hold the read switch and then the range switch. The counter shall conform to the requirements of 3.7.5.3 by the display of the reset indication for approximately 1 second just prior to the display of the range reading.

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4.6.5.4.5 Target discriminator lamps.- Perform this test utilizing the test targets specified in 4.6.5.3.2. The indication of the number of targets shall be displayed by the lighting of the appropriate target discriminator lamp corresponding to the multiple targets observed in the open boresight square of the transceiver when the read switch is depressed. The requirement specified in 3.7.5.5 shall be met.

4.6.5.4.6 Low voltage indicator.- Perform this test with the input voltages to the transceiver adjusted to the specified voltage values indicated in 3.7.5.6. Position the power switch to "ON". Depress the charge switch, the range switch, and the read switch in sequence and observe the numerical indicators. The number "2" shall appear in the unit column in addition to Numbers 0, 1, 5 or 7. The low voltage indicator shall conform to the requirements specified in 3.7.5.6.

4.6.5.5 Performance - transceiver.- Perform the following tests utilizing special testing equipment conforming to 4.5.2.3.4. The level bubble of the adapter mount shall be centered by means of the adapter leveling screws prior to performing any accuracy tests of the transceiver.

4.6.5.5.1 Illumination.- Actuation of the switch on the yoke assembly shall provide illumination to the elevation and azimuth scales and the circular level vial on the mounting adapter, as specified in 3.7.6.1. When the reticle adjustment potentiometer knob of the telescope assembly is rotated in a counterclockwise direction, the reticle light shall conform to the requirements of 3.7.6.1.

4.6.5.5.2 Elevation accuracy.- Perform this test with the elevation scale of the telescope assembly and dial scales of the micrometer of the transceiver indicating the following elevation and depression settings in mils, +75, +150, +300, +500; "0" horizontal line of sight, -75, -150, -300, and -500 mils as specified in 3.7.6.2. With backlash included for the direction of travel of the elevation mechanism, utilizing the elevation knob adjust the transceiver to zero elevation as indicated on the elevation scale of the telescope and dial scales of the micrometer with respect to their zero index without overtravel. Observe through the transceiver eyepiece assembly, the "0" horizontal reticle line of the transceiver shall be in coincidence with respect to the "0" horizontal line of sight of the target image. Continue rotation of the elevation knob in the same direction of travel to elevate the line of sight of the transceiver to each mil increment specified above without overtravel. As each mil increment is established on the elevation and micrometer scales, sight through the transceiver eyepiece assembly. The "0" horizontal reticle line of the transceiver shall be in coincidence with horizontal target images corresponding to the elevation and depression settings introduced on the elevation and micrometer scales. Any displacement errors between the target images with respect to the "0" horizontal reticle line of the transceiver due to elevation change between any 2 target points shall indicate the true position of the line of sight of the transceiver. Record the elevation and depression accuracy readings of the line of sight at each mil increment as measured on the target image. The recorded accuracy readings shall be within the tolerances specified in 3.7.6.2.

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NOTE: Should overtravel occur during rotation of the elevation knob, the test is nullified. The test for elevation accuracy shall be repeated.

4.6.5.5.3 Elevation backlash.- Perform this test with backlash included for the direction of travel of the elevation mechanism. Adjust the transceiver to zero elevation as indicated on the elevation scale and dial scale of the micrometer to their respective zero index without overtravel. Observe through the eyepiece assembly of the transceiver, the "0" horizontal reticle line of the transceiver shall be in coincidence with respect to the "0" horizontal line of sight of the target image. Continue rotation of the elevation knob in the same direction of travel to elevate the line of sight of the transceiver approximately 25 mils, then reverse the direction of travel of the elevation knob with backlash included to return the elevation scale and dial scales to their respective zero index. View through the eyepiece assembly of the transceiver. Any displacement error between the "0" horizontal reticle line with respect to the "0" horizontal target image line shall be recorded as measured on the target image. Backlash readings shall then be made at randomly selected target images (a minimum of 10 positions) throughout +500 mils of elevation excursion using the above test procedure for direction of travel of the elevation mechanism. Record the elevation backlash measurements. Upon completion of the elevation backlash tests, return the transceiver to the "0" horizontal line of sight of the target image. Utilizing the above test procedures, elevation backlash readings shall be performed (a minimum of 10 positions) throughout -500 mils of depression. Record the depression backlash measurements. The recorded backlash measurements shall not exceed the tolerance specified in 3.7.6.3.

NOTE: Should overtravel occur during the rotation of the elevation knob, the test is nullified. The test for elevation backlash shall then be repeated.

4.6.5.5.4 Elevation knob torque.- Standard torque measuring equipment capable of measuring the required torque values shall be utilized for this test. An appropriate adapter to accommodate the elevation knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the knob counterclockwise and clockwise through the full range of required travel. The measured running torque required to rotate the knob in the counterclockwise and clockwise directions shall not exceed the value specified in 3.7.6.4 for the temperatures indicated therein.

4.6.5.5.5 Lateral displacement.- Conduct this test in sequence to the test in 4.6.5.5.3. Elevate the transceiver to zero elevation, as indicated on the elevation scale. Establish coincidence of the vertical reticle line with the vertical target image line. Verify the level of the transceiver by means of the circular level vial bubble. While observing the vertical reticle line of the transceiver, with reference to the vertical target image line, elevate the transceiver in 100 mil increments to a minimum of 500 mils elevation by means of the elevation knob. At each 100 mil increment of elevation, reverse the direction of rotation of the elevation knob by one half a revolution, then continue to the next 100 mil increment of elevation. Any horizontal movement of the vertical reticle line of

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the transceiver as a result of this elevation procedure, shall not exceed the tolerances specified in 3.7.6.5 as measured on the target. Return the transceiver to zero elevation. Depress the line of sight of the transceiver in 100 mil increments for a total of 500 mils utilizing the above procedure. During this 500 mil excursion, there shall be no horizontal displacement of the vertical reticle line in excess of the tolerances specified in 3.7.6.5 as measured on the vertical target image line.

4.6.5.6 Sighting.

4.6.5.6.1 Open sight.- A calibrated wall target capable of measuring the open sight accuracies of the transceiver assembly as specified in 3.7.7.1 shall be utilized with this test. Construction of the wall target shall contain an established level line of sight and a plumb line. While viewing through the telescope eyepiece assembly, the telescope vertical and horizontal reticle lines shall be in coincidence with respect to the plumb line and established level line of sight. A calibrated auxiliary telescope of at least 3 power (3X) containing a vertical and horizontal reticle line shall be positioned on the leveled surface of the testing equipment at the open sight. The horizontal reticle line of the telescope shall be aligned to the horizontal (top) machined surface of the open sight with the vertical reticle line bisecting the vertex of the "V". Observe through the auxiliary telescope and record the position of the intersecting point of the reticle lines with respect to the target image utilized to test the open sight. The intersecting point of the reticle lines shall not exceed the tolerance specified in 3.7.7.1 as measured on the target.

4.6.5.6.2 Collimation (boresight).- Adjust the transceiver to its zero azimuth position as indicated by the microscope reticle and mounting adapter scale. Adjust the transceiver to zero elevation as indicated on the elevation scale of the telescope assembly and dial scale of the elevation knob. Crosslevel the transceiver by means of the mounting adapter leveling screws and circular level vial. Position the Boresight Chart: 10541240 as specified in 3.7.7.2 with its boresight cross aligned to the transceiver reticle. Set the min-range switch to the ".2" position, and the target discriminator switch to the "1" position. Close the parallax shield. Position the power switch to "ON". Fire the laser beam at boresight chart and adjust the transceiver in azimuth and elevation until the laser beam is centered in the rectangle pattern of the target. When the laser beam is centered in the rectangular pattern of the boresight target, the horizontal and vertical reticle line of the transceiver, the elevation and dial scales of the transceiver shall conform to the requirements of 3.7.7.2.

4.6.5.6.3 Boresight adjustment.- Standard torque measuring equipment capable of measuring the required torque values shall be utilized for this test. An appropriate adapter to accommodate the boresight screws shall be used with the torque measuring equipment. The measured torque to rotate the screws shall not exceed the values specified in 3.7.7.3 at the temperatures indicated.

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4.6.5.6.4 Plumb travel.- Conduct this test in sequence to the test in 4.6.5.5.3. Elevate the transceiver 500 mils as indicated on the elevation scale. Establish coincidence of the vertical reticle line of the transceiver with the vertical target image line. Verify the level of the transceiver by means of the circular level vial bubble. Depress the line of sight of the transceiver the required number of mils, by means of the elevation knob, while observing the vertical reticle line with reference to the vertical target image line. During this excursion, the vertical reticle line of the transceiver shall track the target image line within the tolerance specified in 3.7.7.4, as measured on the target.

4.6.5.6.5 Eyepiece focus.- The test shall be made with the aid of a calibrated dioptometer with a magnification of at least 3 power (3X) and a diopter range for measuring the accuracies of the diopter knob scale as specified in 3.7.7.5. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle and set the dioptometer objective scale to zero. Set the diopter scale of the telescope under test to zero diopter. Position the dioptometer at the telescope assembly eyepiece. While viewing through the dioptometer, obtain best focus of the telescope assembly reticle by adjustment of the dioptometer objective scale. The telescope reticle shall be in best focus within the limit specified in 3.7.7.5, as measured on the dioptometer objective scale.

4.6.5.6.6 Diopter adjustment torque.- With an adapter affixed to the diopter knob, rotate the knob through the range specified in 3.7.7.6. The running torque shall not exceed the values specified in 3.7.7.6 for the temperatures indicated.

4.6.5.6.7 Reticle illumination.- Apply the operating voltage to the reticle lamp of the telescope as specified in 3.7.7.7. The illuminated reticle etchings shall be clearly defined when visually examined against a darkened background. Should the brightness of a reticle defect be greater than the reticle lines, the defect shall be cause for rejection. The potentiometer knob shall be rotated clockwise from the extreme counterclockwise position. During this rotation, the potentiometer shall meet the requirements specified in 3.7.7.7 while viewing the reticle pattern.

4.6.5.7 Electromagnetic compatibility.- The transciever shall be subjected to the test methods of MIL-STD-462 for Class IIA equipment for compliance with 3.7.8.

4.6.5.8 Interchangeability.

4.6.5.8.1 Selection.- Three transceivers shall be selected as follows from initial production to determine conformance with 3.7.9:

a. Prior to final assembly, a group of 3 of each of the 14 component modules listed in 3.7.9 shall be presented to the Quality Assurance Representative for selection and final assembly of 3 transceivers.

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b. Transceiver Number 1 shall be assembled by randomly selecting 1 component/module from each group of 3 of the 14 groups.

c. Transceiver Number 2 shall be likewise assembled by randomly selecting 1 component/module of each of the 2 remaining in each of the 14 groups.

d. The remaining 14 component/modules will be assembled as Transceiver Number 3.

4.6.5.8.2 Transceivers.- The 3 transceivers obtained from 4.4.1.2 shall be subjected to all requirements and tests of this specification in compliance with 3.7.9. The selection (4.4.1.2), final assembly and examination and test shall be witnessed by the Quality Assurance Representative. Subsequent to Government approval and acceptance of the first 6 transceivers, 3 additional transceivers shall be selected, assembled and submitted for interchangeability tests in accordance with the above procedures at 25%, 50% and 75% completion of the contract productionand witnessed by the Quality Assurance Representative each time. When the manufacturing process, materials, or inspection techniques previously applied are subject to change, or an adverse change occurs in performance capability or reliability of the transceiver, this shall be cause to cease those operations contributing to the change or failure until corrections have been made by the contractor and approved by the Government.

4.6.5.9 Safety.

4.6.5.9.1 Operation.- Conduct this test in the following sequence:

a. Position the power switch to "ON". Momentarily depress the charge switch and allow a 10 second time period for the pulse forming network to charge. Immediately following the 10 second time period, position the power switch to "OFF". Subsequent to these operations, the pulse forming network and motor of the cavity assembly shall conform to the requirements of 3.7.10.1a, as determined by an audible examination. Discharging of the pulse forming network through the flash lamp will be recongnized as the distinctive popping noise that occurs when the transceiver is fired.

b. Position the power switch to "ON". Depress the range switch without depressing the charge switch. Subsequent to this operation the motor of the cavity assembly shall conform to the requirements of 3.7.10.1b, as determined by an audible examination.

c. Turn the switch of the transceiver to the "ON" position. Press the charge switch to charge the capacitors of the PFNPS. Leaving the switch in the "ON" position, remove the cover of the left housing. (Immediately turn switch to "OFF" position after cover is removed.) The capacitors shall have discharged through the load resistor located in the left housing as indicated by a voltmeter reading of no more than 50 volts when a voltmeter probe is placed on the contact spring of the bracket assembly (10540971) subsequent to separating the spring from contact by an insulator. The requirements of 3.7.10.1c shall be met.

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4.6.5.9.2 Eye filter.- The filter lever of the eye filter assembly shall meet the requirements specified in 3.7.10.2(b) when subjected to the operational temperatures specified in 4.6.2.9. The filter assembly shall be removed and replaced (a minimum of 5 times) from the transceiver line of sight to determine repeatability positioning of the filter assembly and operability of the filter lever.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10540501. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The transceiver is a component of the Range Finder, Laser, AN/GVS-3 and provides the output power, receiving, sighting and ranging functions for the rangefinder.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable packaging data sheet.
c. Level of protection required.
d. Provisions for first article testing.

Custodian:

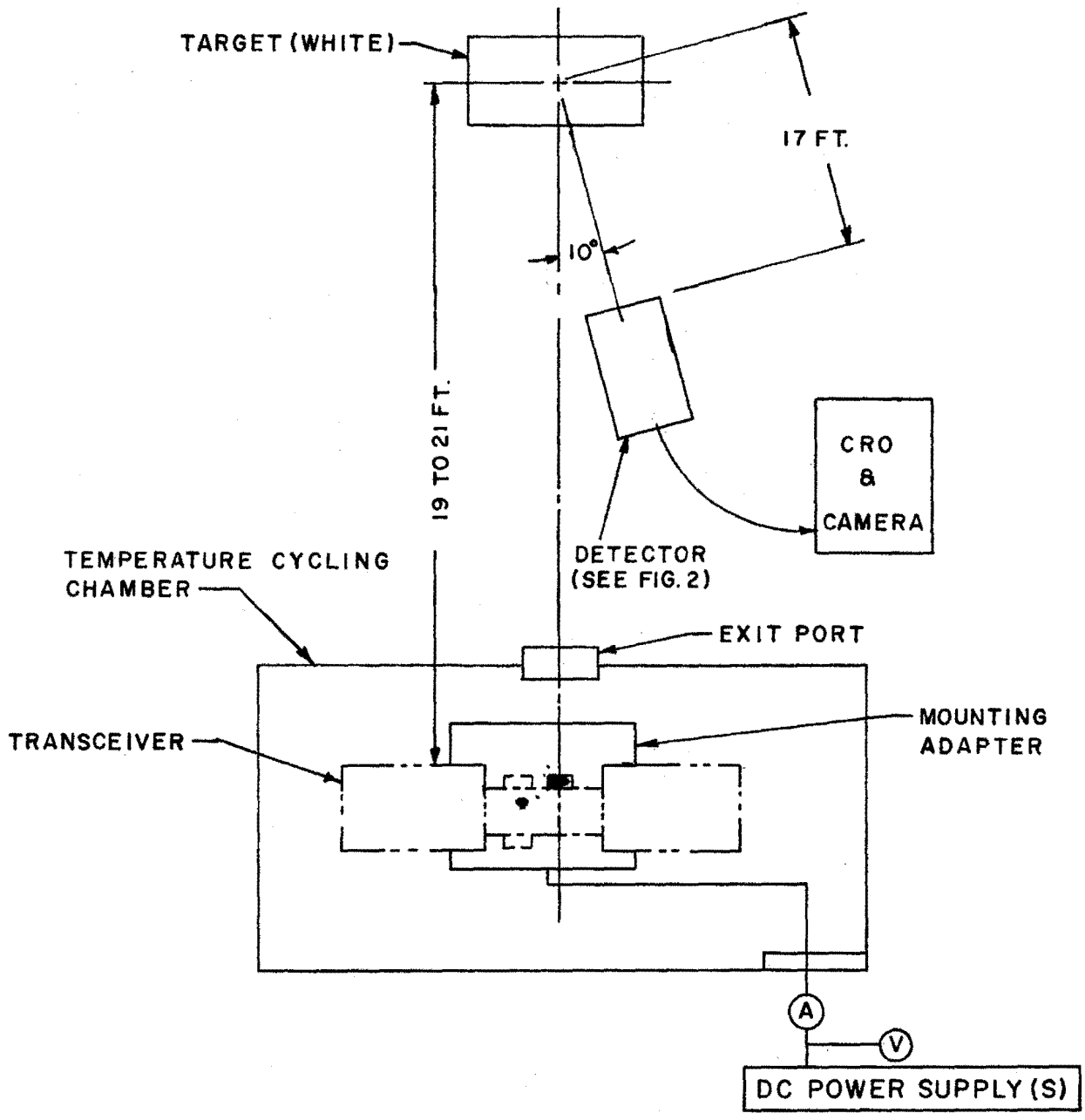
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Preparing activity:

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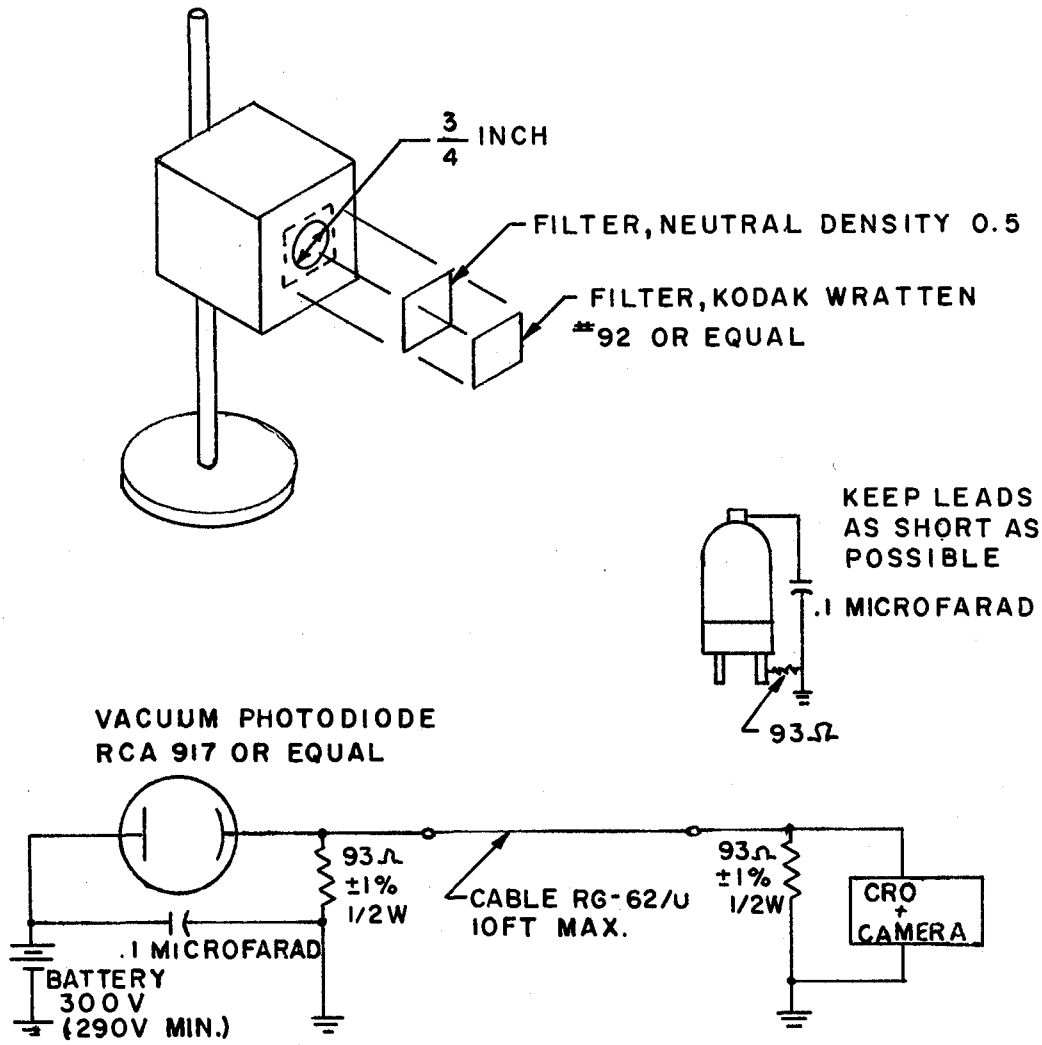


NOTE:
SEE PARAGRAPH 4.5.2.3 AND 4.5.3 FOR INSPECTION EQUIPMENT REQUIREMENTS.

POWER OUTPUT TEST SET-UP

FIG. 1

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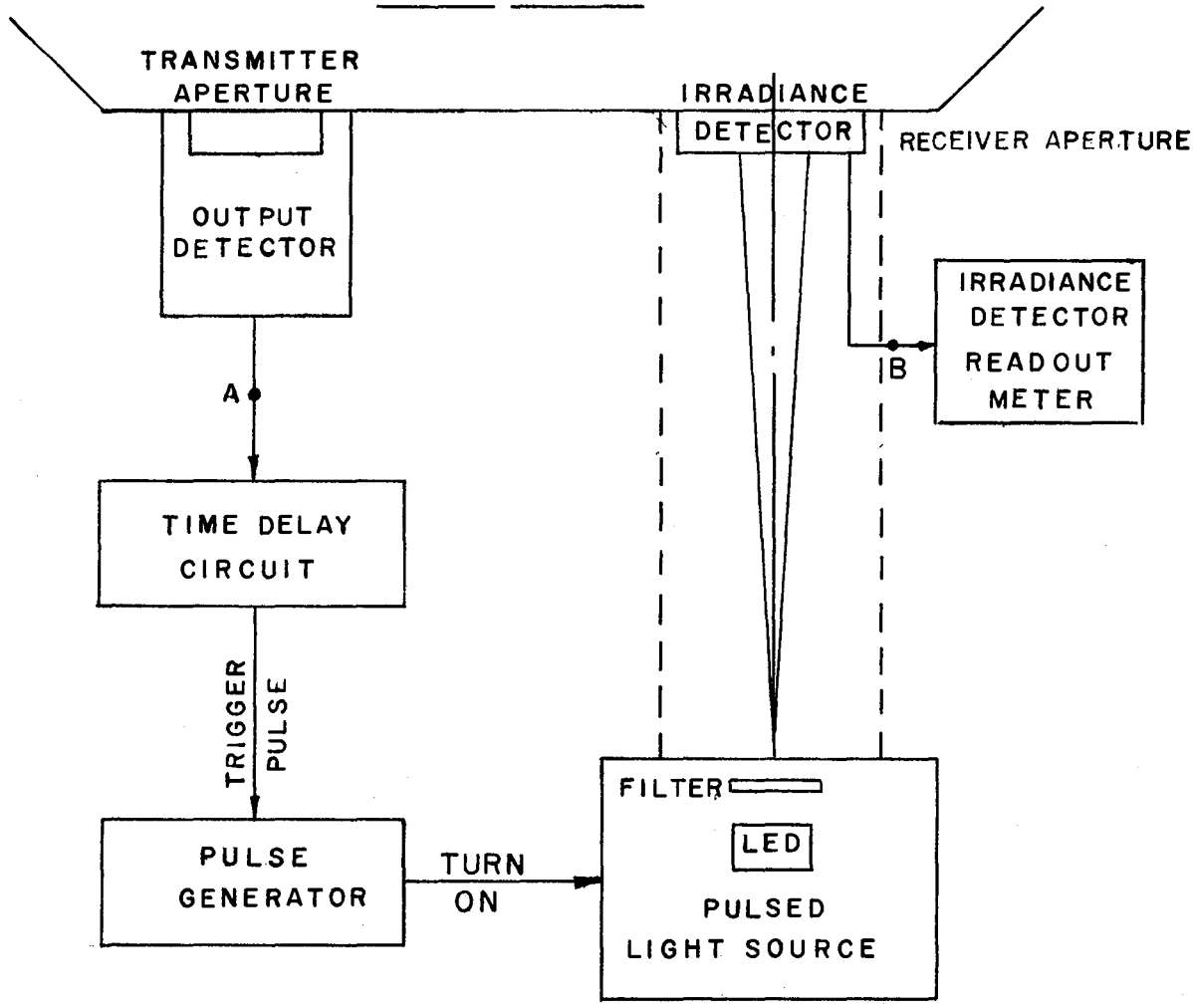


TYPICAL DETECTOR FOR POWER MEASUREMENTS

FIG. 2

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RANGE FINDER.



NOTE: * 1. MEASURE TIME DELAY FROM POINT "A" FROM OUTPUT DETECTOR TO POINT "B" FROM IRRADIANCE DETECTOR.

2. ISOLATE THE RECEIVER APERTURE TO PREVENT ERRONEOUS READINGS FROM REFLECTIONS FROM OUTPUT PULSE.

RECEIVER SENSITIVITY TEST SETUP

FIG. 3

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